STAT 212

Stochastic Processes

Lecture Notes

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Topics:

(i) Advanced Martingales

(ii) Brownian Motion & Stochastic Calculus

(iii) Applications to Generative AI (time permitting)

Grading: Homeworks (20%), Midterm (25%), Final (40%), Project (15%)

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Contents

1 Conditional Expectation **3**
- 1.1 Topics . . . 3
- 1.2 Course Overview and Goals . . . 3
- 1.3 Review (and Generalization) of Conditional Expectation . . . 3
  - 1.3.1 Simple Definition . . . 3
  - 1.3.2 Limitations . . . 4
  - 1.3.3 General Definition of Conditional Expectation . . . 4
- 1.4 Properties of Conditional Expectation . . . 4

2 Conditional Expectation: Existence and Uniqueness, Radon–Nikodym Theorem **6**
- 2.1 Recap from Last Time . . . 6
- 2.2 Uniqueness of Conditional Expectation . . . 6
- 2.3 Absolute Continuity . . . 6
- 2.4 Radon–Nikodym Theorem . . . 7
- 2.5 Existence of Conditional Expectation Assuming Radon–Nikodym Theorem 7
- 2.6 Proof of the Radon–Nikodym Theorem (Proof by Anton Schep, 2003) . . 8

3 Radon–Nikodym Theorem Proof, Martingales **9**
- 3.1 Radon–Nikodym Theorem (continued) . . . 9
  - 3.1.1 Recap . . . 9
  - 3.1.2 Proof that f_* achieves ν . . . 9
- 3.2 Conditional Probability and Conditional Distribution . . . 12
- 3.3 Martingales . . . 12
  - 3.3.1 Setup . . . 12
  - 3.3.2 Submartingale Convergence Theorem . . . 12

4 Uniform Integrability, L^1-Convergence **14**
- 4.1 Agenda . . . 14
- 4.2 Recall: Martingale Convergence . . . 14
- 4.3 Uniform Integrability . . . 15
  - 4.3.1 Exercise: UI Implies L^1-Bounded . . . 15
  - 4.3.2 Characterization of Uniform Integrability . . . 15
- 4.4 Examples of UI Collections . . . 16
- 4.5 Why Care About Uniform Integrability? . . . 17

5 L^1-Convergence (cont.), L^p-Convergence, Doob's Inequalities **19**
5.1 Recall from Last Time . . . 19
5.2 L^1-Convergence of Martingales . . . 19
5.3 L^p-Convergence of Martingales ($p > 1$) . . . 20
5.3.1 Proof of the Theorem Assuming the Lemmas . . . 21
5.4 Doob's Inequalities . . . 21
5.4.1 Proof of Lemma 5.4 . . . 22

6 Reverse Martingales, Hewitt–Savage 0-1 Law, U-Statistics **24**
6.1 Reverse Martingales . . . 24
6.2 Doob's Upcrossing Inequality for Reverse Martingales . . . 25
6.3 Boosting to L^1-Convergence . . . 25
6.4 Showing $X_{-\infty} = \mathbb{E}[X_0 \mid \mathcal{F}_{-\infty}]$ . . . 25
6.5 Application: Convergence of U-Statistics of i.i.d. Random Variables . . . 26

7 SLLN with Reverse Martingales, Brownian Motion **28**
7.1 Brownian Motion . . . 28
7.1.1 Historical Background . . . 28
7.1.2 Definition of Standard Brownian Motion . . . 28
7.1.3 Equivalent Definitions . . . 29
7.1.4 Existence and Uniqueness . . . 29
7.2 Uniqueness of Brownian Motion . . . 29

8 Existence of Brownian Motion, Scaling Properties **31**
8.1 Recap . . . 31
8.2 Existence of Brownian Motion (Paul Lévy's Construction) . . . 31
8.3 Brownian Motion on $[0, \infty)$ . . . 33
8.3.1 Construction and Uniqueness . . . 33
8.4 Scaling Properties of Brownian Motion . . . 34

9 Properties of Brownian Motion **36**
9.1 Almost Sure Nowhere Differentiability of Brownian Motion . . . 36
9.2 Markov Property of Brownian Motion . . . 38
9.3 Filtrations . . . 38

10 Markov Property, Stopping Times, Strong Markov Property **39**
10.1 Right-Continuous Filtrations . . . 39
10.2 Markov Property with $\mathcal{F}_t^+$ . . . 39
10.3 Stopping Times . . . 40
10.4 Blumenthal's 0/1 Law . . . 40
10.5 Strong Markov Property . . . 41
10.6 Application: Running Maximum is Not a Stopping Time . . . 42

11 Reflection Principle, Continuous-Time Martingales **43**
11.1 Reflection Principle for Brownian Motion . . . 43
11.2 Distribution of the Running Maximum . . . 44
11.3 Continuous-Time Martingales . . . 44

Chapter 1

Conditional Expectation

1.1 Topics

1. Conditional Expectation.
2. Radon–Nikodym Theorem.

1.2 Course Overview and Goals

Q. What is this course about?

Stochastic Processes $\{X_t : t \in \mathbb{T}\}$, $X_t \in \mathcal{S}$ — state space, $\mathbb{T}$ — index set.

Q. Why care about stochastic processes? — To model & predict dynamics occurring in the physical world, e.g. weather dynamics, infection counts, stock prices, etc.

Interest: Can we predict future behavior? Can we use dynamics to produce "samples" of interest? — applications in Gen AI.

Our approach in this course: Focus on Brownian Motion & allied stochastic processes.

Why? BM is a canonical stochastic process with many fascinating properties. Using Itô calculus, we can construct many other stochastic properties from BM.

Finally: The techniques & principles we learn can be widely used to study other stochastic processes.

Before diving into a systematic study of BM, we have to build up some technical foundations.

1.3 Review (and Generalization) of Conditional Expectation

1.3.1 Simple Definition

Definition 1.1. Conditional Expectation (Simple)

Let X, Y be random variables with $\mathbb{E}\,|Y| < \infty$. The *conditional expectation* $\mathbb{E}[Y \mid X]$ is the (almost surely) unique function $g(X)$ that uncorrelates $Y - g(X)$ from $h(X)$

for all bounded, measurable $h\colon \mathbb{R} \to \mathbb{R}$. Formally,

$$\mathbb{E}\Big[(Y - g(X))h(X)\Big] = 0.$$

1.3.2 Limitations

(i) The previous definition works if $X \in \mathbb{R}^n$ for some $n \geq 1$. In working with continuous-time stochastic processes, we want to define quantities like $\mathbb{E}\Big[X_{t+h} \mid \{X_s : s \leq t\}\Big]$.

Q. How to make sense of this?

(ii) Existence & uniqueness of stochastic conditional expectations? (Can use L^2-projection if $\mathbb{E}Y^2 < \infty$, but general construction is more involved.)

1.3.3 General Definition of Conditional Expectation

Definition 1.2. Conditional Expectation (General)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. $X\colon (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ is measurable and $\mathbb{E}\,|X| < \infty$. Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-field.
Then $\mathbb{E}[X \mid \mathcal{G}]$ is the unique (up to measure zero) $\mathcal{G}$-measurable function such that

$$\mathbb{E}[X\mathbb{1}_A] = \mathbb{E}\Big[\mathbb{E}[X \mid \mathcal{G}]\mathbb{1}_A\Big] \quad \forall\, A \in \mathcal{G}.$$

Remark 1.3

(1) This definition generalizes the simple definition (Definition 1.1), once we take $\mathcal{G} = \sigma(X)$.

(2) For continuous-time stochastic processes, we can work with

$$\mathbb{E}\Big[X_{t+h} \mid \sigma(\{X_s : s \leq t\})\Big].$$

Q. Existence & uniqueness? — *We will come back to this.*

1.4 Properties of Conditional Expectation

Theorem 1.4. Properties of Conditional Expectation

Let X, Y satisfy $\mathbb{E}\,|X|, \mathbb{E}\,|Y| < \infty$.

(a) **(Linearity)** $\mathbb{E}[aX + Y \mid \mathcal{G}] = a\,\mathbb{E}[X \mid \mathcal{G}] + \mathbb{E}[Y \mid \mathcal{G}]$.

(b) **(Monotonicity)** $X \leq Y \implies \mathbb{E}[X \mid \mathcal{G}] \leq \mathbb{E}[Y \mid \mathcal{G}]$.

(c) **(Monotone Convergence)** $X_n \geq 0$, $X_n \uparrow X$, $\mathbb{E}X < \infty$

$$\implies \mathbb{E}[X_n \mid \mathcal{G}] \uparrow \mathbb{E}[X \mid \mathcal{G}].$$

Proof. See Theorem 4.1.9 (Durrett). □

Theorem 1.5. Jensen's Inequality for Conditional Expectation

Let $\varphi\colon \mathbb{R} \to \mathbb{R}$ be convex. Then

$$\mathbb{E}[\varphi(X) \mid \mathcal{G}] \geq \varphi\Big(\mathbb{E}[X \mid \mathcal{G}]\Big).$$

Proof sketch. (See Durrett Theorem 4.1.10 for full proof.)
Assume φ is differentiable and $\mathbb{E}\Big[|\varphi'(\mathbb{E}[X \mid \mathcal{G}])|\Big] < \infty$ (or $|\varphi'(x)| \leq 1$ for all x). By convexity,

$$\varphi(X) \geq \varphi\Big(\mathbb{E}[X \mid \mathcal{G}]\Big) + \varphi'\Big(\mathbb{E}[X \mid \mathcal{G}]\Big)\Big(X - \mathbb{E}[X \mid \mathcal{G}]\Big).$$

Taking conditional expectations,

$$\mathbb{E}[\varphi(X) \mid \mathcal{G}] \geq \varphi\Big(\mathbb{E}[X \mid \mathcal{G}]\Big).$$

Extending to the general case requires some properties of convex functions. □

Corollary 1.6

(1) $|\mathbb{E}[X \mid \mathcal{G}]| \leq \mathbb{E}[|X| \mid \mathcal{G}]$. — Use Jensen's inequality on $\varphi(x) = |x|$.

(2) $X_1, \ldots, X_n \xrightarrow{L^1} X \implies \mathbb{E}[X_n \mid \mathcal{G}] \xrightarrow{L^1} \mathbb{E}[X \mid \mathcal{G}]$.

Proof. For (2): $X_n \xrightarrow{L^1} X \implies \mathbb{E}\Big[|X - X_n| \mid \mathcal{G}\Big] \xrightarrow{L^1} 0 \implies \mathbb{E}[X_n \mid \mathcal{G}] \xrightarrow{L^1} \mathbb{E}[X \mid \mathcal{G}]$. □

Chapter 2

Conditional Expectation: Existence and Uniqueness, Radon–Nikodym Theorem

Agenda

(i) Conditional Expectation: Existence & Uniqueness.

(ii) Radon–Nikodym Theorem.

2.1 Recap from Last Time

Last time: $(\Omega, \mathcal{F}, \mathbb{P})$ probability space. $\mathcal{G} \subseteq \mathcal{F}$ sub σ-field. $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ measurable with $\mathbb{E}\,|X| < \infty$.

The conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the unique (up to measure zero sets) $\mathcal{G}$-measurable random variable such that

$$\forall\, A \in \mathcal{G}, \quad \mathbb{E}[X \mathbb{1}_A] = \mathbb{E}\Big[\mathbb{E}[X \mid \mathcal{G}]\, \mathbb{1}_A\Big].$$

2.2 Uniqueness of Conditional Expectation

Uniqueness: If possible, let Y_1 and Y_2 be versions of $\mathbb{E}[X \mid \mathcal{G}]$.

Let $A = \{Y_1 > Y_2\} \in \mathcal{G}$.

$$\mathbb{E}[Y_1 \mathbb{1}_A] = \mathbb{E}[X \mathbb{1}_A] = \mathbb{E}[Y_2 \mathbb{1}_A] \quad \Rightarrow \quad \mathbb{E}\Big[(Y_1 - Y_2)\mathbb{1}_A\Big] = 0.$$

$$\Rightarrow \quad \mathbb{P}(A) = 0.$$

Similarly, $\mathbb{P}(Y_2 > Y_1) = 0$.

$$\Rightarrow \quad \mathbb{P}(Y_1 \neq Y_2) = 0.$$

To get existence of $\mathbb{E}[X \mid \mathcal{G}]$, we will use the **Radon–Nikodym theorem**.

2.3 Absolute Continuity

$(\Omega, \mathcal{F})$ is a measurable space. μ, ν are finite measures on $(\Omega, \mathcal{F})$.

Definition 2.1. Absolute Continuity

ν is *absolutely continuous* with respect to μ (written $\nu \ll \mu$) if

$$\mu(S) = 0 \quad \Rightarrow \quad \nu(S) = 0.$$

2.4 Radon–Nikodym Theorem

Theorem 2.2. Radon–Nikodym Theorem

(a) If $\nu \ll \mu$, there exists $f \geq 0$, $\int f \, d\mu < \infty$ such that

$$\nu(S) = \int_S f \, d\mu \qquad \forall \, S \in \mathcal{F}.$$

Denote $f = \dfrac{d\nu}{d\mu}$.

(b) Without absolute continuity, there exist $f \geq 0$, $\int f \, d\mu < \infty$, and a finite measure $\oplus$ such that

$$\nu(S) = \oplus(S) + \int_S f \, d\mu.$$

Further, $\oplus$ and μ have disjoint support, i.e.,

$$\exists \, S \in \mathcal{F}, \quad \mu(S) = 0, \quad \oplus(S^c) = 0.$$

$\oplus$ is the "singular part" and the second term is the "absolutely continuous part" of ν.

(c) Assume $\nu(S) \leq \mu(S)$ for all $S \in \mathcal{F}$. Then there exists $f : \Omega \to [0, 1]$ measurable such that

$$\nu(S) = \int_S f \, d\mu.$$

Remark 2.3. Intuition for the Radon–Nikodym derivative

The RN derivative is reweighing of μ.

2.5 Existence of Conditional Expectation Assuming Radon–Nikodym Theorem

$X : \Omega \to \mathbb{R}$ with $\mathbb{E}\,|X| < \infty$. $\mathcal{G} \subseteq \mathcal{F}$ sub σ-field.

Define: $\nu(S) = \int_S X \, d\mu$ (Assume $X \geq 0$. Otherwise $X = X_+ - X_-$ and work with X_+ and X_- separately) for $S \in \mathcal{G}$.

$\mathbb{E}\,|X| < \infty$ implies ν is a finite measure.

If $\mu(S) = 0 \Rightarrow \nu(S) = 0$.

By RN theorem, there exists $Y = \dfrac{d\nu}{d\mu}$ which is $\mathcal{G}$-measurable.

Further,

$$\nu(S) = \int_S Y \, d\mu = \int_S X \, d\mu \qquad \forall \, S \in \mathcal{G}.$$

$$\Rightarrow \quad Y = \mathbb{E}[X \mid \mathcal{G}].$$

2.6 Proof of the Radon–Nikodym Theorem (Proof by Anton Schep, 2003)

For two general finite measures μ, ν.

Obs: (b) $\Rightarrow$ (a) $\Rightarrow$ (c). For this pair to get (a), (b) (HW), will use (c) on $\nu \leq \nu + \mu$. Will prove (c).

Main challenge: Need to come up with the RN derivative out of nowhere.

Idea: Look at

$$\mathcal{H} = \left\{ f : \Omega \to [0,1] \text{ measurable}, \quad \int_S f \, d\mu \leq \nu(S) \quad \forall\, S \in \mathcal{F} \right\}.$$

$\dfrac{d\nu}{d\mu}$ should be the maximal element in this class.

Proof. **Step 0:** $\mathcal{H} \neq \varnothing$, since $f = 0 \in \mathcal{H}$.

Step 1: If $f_1, f_2 \in \mathcal{H}$, then $\max\{f_1, f_2\} \in \mathcal{H}$.
Fix $S \in \mathcal{F}$.

$$\begin{aligned} \int_S \max\{f_1, f_2\} \, d\mu &= \int_{S \cap \{f_1 \geq f_2\}} f_1 \, d\mu + \int_{S \cap \{f_1 < f_2\}} f_2 \, d\mu \\ &\leq \nu\Big(S \cap \{f_1 \geq f_2\}\Big) + \nu\Big(S \cap \{f_1 < f_2\}\Big) = \nu(S). \end{aligned}$$

Step 2: Define

$$M = \sup_{f \in \mathcal{H}} \int_\Omega f \, d\mu \leq 1.$$

For each $n \geq 1$, there exists $f_n \in \mathcal{H}$ such that

$$\int_\Omega f_n \, d\mu \geq M - \tfrac{1}{n}.$$

Set $\bar{f}_n = \max\{f_1, \ldots, f_n\} \in \mathcal{H}$.
$\bar{f}_n \uparrow$ pointwise and bounded by 1.

$$\boxed{f_* = \lim_{n \to \infty} \bar{f}_n}$$

Step 3: $\displaystyle\int_\Omega f_* \, d\mu = M$.

By definition, $\displaystyle\int_\Omega f_* \, d\mu \leq M$ (since $f_* \in \mathcal{H}$, by Fatou's lemma).
However,

$$\int_\Omega f_* \, d\mu \geq \lim_{n \to \infty} \int_\Omega \bar{f}_n \, d\mu = M$$

(pointwise convergence, Fatou's lemma).

Step 4: $\forall\, S \in \mathcal{F}, \quad \displaystyle\int_S f_* \, d\mu \leq \lim \int_S \bar{f}_n \, d\mu \leq \nu(S) \quad$ (since $\bar{f}_n \in \mathcal{H}$).

Step 5: To show $\displaystyle\int_S f_* \, d\mu = \nu(S) \quad \forall\, S \in \mathcal{F}$. $\square$

Chapter 3

Radon–Nikodym Theorem Proof, Martingales

Agenda

1. Radon–Nikodym Theorem
2. Martingales

Announcement. Lectures in G125 from now on.

3.1 Radon–Nikodym Theorem (continued)

3.1.1 Recap

So far: $(\Omega, \mathcal{F}, \mathbb{P})$ is a measure (probability) space. μ, ν are finite measures on $(\Omega, \mathcal{F})$, with $\nu(S) \leq \mu(S)$ for all $S \in \mathcal{F}$.

To prove: There exists $f : \Omega \to [0, 1]$ such that

$$\nu(S) = \int_S f \, d\mu.$$

Have done: Define

$$\mathcal{H} = \left\{ f : \Omega \to [0,1] \text{ measurable} \;\middle|\; \int_S f \, d\mu \leq \nu(S) \;\; \forall\, S \in \mathcal{F} \right\}, \qquad M = \sup_{f \in \mathcal{H}} \int f \, d\mu.$$

Then there exists $f_* \in \mathcal{H}$ such that $\int f_* \, d\mu = M$. (Recall: f_* was constructed as the pointwise limit of $\bar{f}_n$, where $\bar{f}_n = \max(f_1, \ldots, f_n)$ for a maximizing sequence $\{f_n\}$ in $\mathcal{H}$.)

3.1.2 Proof that f_* achieves ν

Claim 3.1. f_* **satisfies** $\nu(S) = \int_S f_* \, d\mu$

For every $S \in \mathcal{F}$,

$$\nu(S) = \int_S f_* \, d\mu.$$

Proof. Suppose not. Then there exists $\bar{S} \in \mathcal{F}$ such that

$$\nu(\bar{S}) > \int_{\bar{S}} f_* \, d\mu.$$

Strategy: Use $\bar{S}$ to construct $\hat{S} \in \mathcal{F}$ with $\mu(\hat{S}) > 0$ such that $f_* + \varepsilon \mathbb{1}_{\hat{S}} \in \mathcal{H}$. But then

$$\int (f_* + \varepsilon \mathbb{1}_{\hat{S}}) \, d\mu > M,$$

which is a contradiction.

Construction strategy: Iterative construction, carving out bad sets at each step.

Step 1: Observations.

(i) *Observation on $\bar{S}$:* Since $\nu \leq \mu$ (by definition) and $f_* \in \mathcal{H}$,

$$\nu\Big(S \cap \{f_* = 1\}\Big) \leq \mu\Big(S \cap \{f_* = 1\}\Big), \quad \text{(by definition, } \nu \leq \mu\text{)},$$

and

$$\nu\Big(S \cap \{f_* = 1\}\Big) \geq \int_{S \cap \{f_* = 1\}} f_* \, d\mu = \mu\Big(S \cap \{f_* = 1\}\Big).$$

So the deficit $\nu(\bar{S}) - \int_{\bar{S}} f_* \, d\mu$ is concentrated on $\bar{S} \cap \{f_* < 1\}$.

(ii) *Shrinking to S_0:* Start with $S_0 = \bar{S} \setminus \{f_* = 1\}$. Let $F_n = \{f_* \leq 1 - \frac{1}{n}\}$. Then $F_n \uparrow \{f_* < 1\}$.

Since $\int_{\bar{S}} f_* \, d\mu < \nu(\bar{S})$, there exists n_0 large enough such that

$$\int_{S \cap F_n} f_* \, d\mu < \nu(S \cap F_n), \qquad \forall \, n \geq n_0.$$

So there exists n large enough (set $\varepsilon = \varepsilon(n)$) such that:

(a) $f_* + \varepsilon \mathbb{1}_{S \cap F_n} \leq 1$ on $S \cap F_n$ (since $f_* \leq 1 - \frac{1}{n}$ on F_n, choosing $\varepsilon \leq \frac{1}{n}$),

(b) $\int_{S \cap F_n} (f_* + \varepsilon \mathbb{1}_{S \cap F_n}) \, d\mu < \nu(S \cap F_n)$.

Setting $\bar{S} = S \cap F_n$, the function $f_* + \varepsilon \mathbb{1}_{\bar{S}}$ seems like a good start.

Issue: Is $f_* + \varepsilon \mathbb{1}_{\bar{S}} \in \mathcal{H}$? We need $\int_T (f_* + \varepsilon \mathbb{1}_{\bar{S}}) \, d\mu \leq \nu(T)$ for all $T \in \mathcal{F}$, not just $T = \bar{S}$.

Step 2: Shrinking $\bar{S}$ to $\hat{S}$.

Idea: Need to "shrink" $\bar{S}$ to $\hat{S}$ so that $f_* + \varepsilon \mathbb{1}_{\hat{S}} \in \mathcal{H}$, i.e.,

$$\int_S (f_* + \varepsilon \mathbb{1}_{\hat{S}}) \, d\mu \leq \nu(S), \qquad \forall \, S \in \mathcal{F}.$$

Observation: If $T \subseteq \bar{S}^c$, then

$$\int_T (f_* + \varepsilon \mathbb{1}_{\bar{S}}) \, d\mu \leq \nu(T),$$

since the indicator does not matter there. So we need to worry only about $T \subseteq \bar{S}$.

Step 3: The Deficit.

Definition 3.2. Deficit

For $A \in \mathcal{F}$, define

$$\mathrm{Def}_\varepsilon(A) = \nu(A) - \int_A (f_* + \varepsilon \mathbb{1}_A)\, d\mu.$$

Note that Def_ε is additive (as a function of A, for sets contained in $\bar{S}$).

By construction, $\mathrm{Def}_\varepsilon(\bar{S}) > 0$.

Idea: If $T \subseteq \bar{S}$ violates the deficit, i.e., $\mathrm{Def}_\varepsilon(T) < 0$, then

$$\mathrm{Def}_\varepsilon(\bar{S} \setminus T) = \mathrm{Def}_\varepsilon(\bar{S}) - \mathrm{Def}_\varepsilon(T) > 0.$$

So: we will progressively rule out the bad sets.

Step 4: Progressive elimination of bad sets.
Define

$$a_k = \inf_{S_k} \mathrm{Def}_\varepsilon(S_k), \qquad S_k \subseteq \bar{S},$$

where S_k is chosen to be disjoint from the union of $S_1, \ldots, S_{k-1}$.
Pick S_k such that $\mathrm{Def}_\varepsilon(S_k) \leq a_k + \frac{1}{k}$. (Observe: $a_k \leq 0$ since $\emptyset$ is always allowed.)
Define:

$$\hat{S} = \bar{S} \setminus \bigcup_{k \geq 1} S_k.$$

Then $\mathrm{Def}_\varepsilon(\hat{S}) > 0$, which implies $\nu(\hat{S}) > 0$, hence $\mu(\hat{S}) > 0$.

Step 5: Verifying $f_* + \varepsilon \mathbb{1}_{\hat{S}} \in \mathcal{H}$.

Claim 3.3. $f_* + \varepsilon \mathbb{1}_{\hat{S}} \in \mathcal{H}$

The function $f_* + \varepsilon \mathbb{1}_{\hat{S}}$ belongs to $\mathcal{H}$.

Suppose not. Then there exists $T \subseteq \hat{S}$ such that

$$\int_T (f_* + \varepsilon \mathbb{1}_{\hat{S}})\, d\mu > \nu(T) + \delta$$

for some $\delta > 0$, i.e., $\mathrm{Def}_\varepsilon(T) < -\delta$.
Since $T \subseteq \hat{S}$, we have $T \cap S_k = \emptyset$ for all $k \geq 1$. So T could have been used as S_k for $k > 1/\delta$.
But then for k large enough,

$$S_k \cup T \implies \mathrm{Def}_\varepsilon(T \cup S_k) \leq \mathrm{Def}_\varepsilon(S_k) - \delta \leq a_k + \frac{1}{k} - \delta < a_k,$$

where the last inequality holds for $k > 1/\delta$. This is a contradiction with the definition of a_k as the infimum!

Therefore $f_* + \varepsilon \mathbb{1}_{\hat{S}} \in \mathcal{H}$, and

$$\int (f_* + \varepsilon \mathbb{1}_{\hat{S}})\, d\mu > M,$$

contradicting the definition of M. This completes the proof. $\square$

3.2 Conditional Probability and Conditional Distribution

So far: $(\Omega, \mathcal{F}, \mathbb{P})$ probability space, $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-field. If $\mathbb{E}|X| < \infty$, we can define $\mathbb{E}[X \mid \mathcal{G}]$.

Q. What about conditional probability? We can use $\mathbb{E}[\mathbb{1}_A \mid \mathcal{G}]$.

Q. What about conditional distribution? Same idea, but need to worry a bit more about measure-zero sets. (More in Sections.)

3.3 Martingales

3.3.1 Setup

$(\Omega, \mathcal{F}, \mathbb{P})$ probability space. $\{\mathcal{F}_n : n \geq 1\}$ is a *filtration*:

- $\mathcal{F}_n$ is a σ-algebra, $\mathcal{F}_n \subseteq \mathcal{F}$,
- $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$.

$\{X_n : n \geq 1\}$ is *adapted* to $\mathcal{F}_n$, i.e.,

$$\{X_n \in A\} \in \mathcal{F}_n.$$

Definition 3.4. Martingale, Sub-martingale, Super-martingale

We say $\{(X_n, \mathcal{F}_n) : n \geq 1\}$ is a

(i) **martingale** if $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$,

(ii) **sub-martingale** if $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n$,

(iii) **super-martingale** if $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n$.

Example 3.5. Random walk martingale

Let ε_i be i.i.d. with $\mathbb{E}[\varepsilon_i] = 0$ and $\mathbb{E}[\varepsilon_i^2] = 1$. Define

$$S_k = \varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_k.$$

Then $\{S_k\}$ is a martingale with respect to $\mathcal{F}_n = \sigma(\{S_0, S_1, \ldots, S_n\})$.

3.3.2 Submartingale Convergence Theorem

Theorem 3.6. Submartingale Convergence Theorem

If $\{(X_n, \mathcal{F}_n) : n \geq 1\}$ is a sub-martingale with

$$\sup_{n \geq 1} \mathbb{E}[X_n^+] < \infty,$$

then $X_n \xrightarrow{\text{a.s.}} X_\infty$.

Q. What can we say about the limit? For example, what about $\mathbb{E}[X_\infty]$? $\mathbb{E}[X_\infty^m]$?

Naive hope: If $\{X_n : n \geq 1\}$ is a martingale, then $\mathbb{E}[X_n] = \mathbb{E}[X_1]$ for all $n \geq 1$. Maybe $\mathbb{E}[X_n] = \mathbb{E}[X_\infty]$?

Unfortunately no! Will need additional conditions. Next time.

Chapter 4

Uniform Integrability, L^1-Convergence

4.1 Agenda

1. Uniform Integrability.
2. L^1-convergence of martingales.

4.2 Recall: Martingale Convergence

Recall the *Martingale Convergence Theorem*: if $\{(X_n, \mathcal{F}_n) : n \geq 1\}$ is a martingale, set

$$M = \sup_{n \geq 1} \mathbb{E}\,|X_n| \quad (\leq \infty).$$

Define

$$X_\infty(\omega) = \begin{cases} \lim_{n\to\infty} X_n(\omega) & \text{if the limit exists,} \\ 0 & \text{otherwise.} \end{cases}$$

Theorem 4.1. L^1-Convergence of Martingales (TFAE)

The following are equivalent:

(i) $M < \infty$ and $X_n \xrightarrow{\text{a.s.}} X_\infty$.

(ii) $\mathbb{E}[X_\infty \mid \mathcal{F}_n] = X_n$.

(iii) $M < \infty$ and $\mathbb{E}\,|X_\infty| = \lim_{n\to\infty} \mathbb{E}\,|X_n|$.

(iv) $\{X_n : n \geq 1\}$ is uniformly integrable.

Q. What is uniform integrability?

4.3 Uniform Integrability

Definition 4.2. Uniform Integrability

A collection $S \subseteq L^1(\Omega, \mathcal{F}, \mathbb{P})$ is *uniformly integrable* (UI) if for every $\varepsilon > 0$, there exists $c > 0$ such that

$$\sup_{X \in S} \mathbb{E}\Big[|X| \, \mathbb{1}(|X| > c)\Big] < \varepsilon.$$

Note: the same c works for all $X \in S$.

Remark 4.3

For any single $X \in L^1$, we have $\mathbb{E}[|X| \, \mathbb{1}(|X| > c)] \to 0$ as $c \to \infty$ (by the dominated convergence theorem). Uniform integrability requires this convergence to be *uniform* over S.

4.3.1 Exercise: UI Implies L^1-Bounded

Exercise. If $S \subseteq L^1(\Omega, \mathcal{F}, \mathbb{P})$ is UI, then S is L^1-bounded (i.e., $\sup_{X \in S} \mathbb{E}\,|X| < \infty$).
Give a counterexample to show that the opposite implication is false.
Q. Which L^1-bounded subsets are UI?

4.3.2 Characterization of Uniform Integrability

Lemma 4.4. Characterization of UI

$S \subseteq L^1(\Omega, \mathcal{F}, \mathbb{P})$ is UI if and only if

(a) S is L^1-bounded, and

(b) (*uniform absolute continuity*) for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $f \in S$ and all $A \in \mathcal{F}$ with $\mathbb{P}(A) < \delta$,

$$\int_A |f| \, d\mathbb{P} < \varepsilon.$$

Proof. **(UI $\implies$ $(a) + (b)$):**
First, UI $\implies$ L^1-bounded, giving (a).
For (b): let $\varepsilon > 0$. We have

$$\mathbb{E}\Big[|f| \, \mathbb{1}_A\Big] = \mathbb{E}\Big[|f| \, \mathbb{1}_{\{|f| \le c\} \cap A}\Big] + \mathbb{E}\Big[|f| \, \mathbb{1}_{\{|f| > c\} \cap A}\Big] \le c \cdot \delta + \sup_{f \in S} \mathbb{E}\Big[|f| \, \mathbb{1}_{\{|f| > c\}}\Big].$$

Choose c such that the second term (using UI) is $< \varepsilon/2$. Then choose δ such that $c \cdot \delta < \varepsilon/2$, giving RHS $< \varepsilon$.

($(a) + (b)$ $\implies$ UI):
Since S is L^1-bounded,

$$\mathbb{P}\Big(|f| > c\Big) \le \frac{\sup_{f \in S} \mathbb{E}\,|f|}{c} < \delta$$

for c large enough. Then by (b),

$$\mathbb{E}\Big[|f| \, \mathbb{1}_{\{|f| > c\}}\Big] < \varepsilon. \qquad \square$$

4.4 Examples of UI Collections

Lemma 4.5. Conditional Expectations Are UI

Let $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$. Define

$$S = \Big\{\mathbb{E}[X \mid \mathcal{G}] : \mathcal{G} \subseteq \mathcal{F} \text{ is a sub-}\sigma\text{-algebra}\Big\}.$$

Then S is UI.

Proof. First, S is L^1-bounded since by Jensen's inequality,

$$\mathbb{E}\,|\mathbb{E}[X \mid \mathcal{G}]| \leq \mathbb{E}\,|X|\,.$$

Next, by Jensen's inequality,

$$\mathbb{E}\Big[|\mathbb{E}[X \mid \mathcal{G}]|\,\mathbb{1}\Big(|\mathbb{E}[X \mid \mathcal{G}]| \geq c\Big)\Big] \leq \mathbb{E}\Big[|X|\,\mathbb{1}\Big(|\mathbb{E}[X \mid \mathcal{G}]| \geq c\Big)\Big].$$

But

$$\mathbb{P}\Big(|\mathbb{E}[X \mid \mathcal{G}]| \geq c\Big) \leq \frac{\mathbb{E}\,|X|}{c} < \delta$$

for c large enough. So

$$\sup_S \mathbb{E}\Big[|\mathbb{E}[X \mid \mathcal{G}]|\,\mathbb{1}\Big(|\mathbb{E}[X \mid \mathcal{G}]| \geq c\Big)\Big] \leq \sup_S \mathbb{E}\Big[|X|\,\mathbb{1}\Big(|\mathbb{E}[X \mid \mathcal{G}]| \geq c\Big)\Big].$$

Claim: RHS $\to 0$ as $c \to \infty$.
Suppose not. Then there exist $\mathcal{G}_n$ and $c_n \uparrow \infty$ such that

$$\mathbb{E}\Big[|X|\,\mathbb{1}\Big(|\mathbb{E}[X \mid \mathcal{G}_n]| \geq c_n\Big)\Big] > \varepsilon.$$

Let $Y_n = |X|\,\mathbb{1}\Big(|\mathbb{E}[X \mid \mathcal{G}_n]| \geq c_n\Big) \leq |X|$. Since $\mathbb{E}\,|X| < \infty$ and $Y_n \xrightarrow{p} 0$ (because $\mathbb{P}(|\mathbb{E}[X \mid \mathcal{G}_n]| \geq c_n) \to 0$), the dominated convergence theorem gives $\mathbb{E}[Y_n] \to 0$, a contradiction. □

Lemma 4.6. L^p-Bounded Implies UI

If $S \subseteq L^1(\Omega, \mathcal{F}, \mathbb{P})$ is L^p-bounded for some $p > 1$ (i.e., $\sup_{X \in S} \mathbb{E}\,|X|^p < \infty$), then S is UI.

Proof. By the Cauchy–Schwarz inequality (more precisely, Hölder's inequality with exponents p and $p/(p-1)$, applied to the case $p = 2$ for simplicity):

$$\mathbb{E}\Big[|X|\,\mathbb{1}(|X| > c)\Big] \leq \sqrt{\mathbb{E}X^2 \cdot \mathbb{P}(|X| > c)} \leq \sqrt{M^2 \cdot \frac{\mathbb{E}X^2}{c}} = \frac{M^2}{\sqrt{c}} \to 0$$

as $c \to \infty$, where $M^2 = \sup_{X \in S} \mathbb{E}X^2$.
More generally, for $p > 1$:

$$\mathbb{E}\Big[|X|\,\mathbb{1}(|X| > c)\Big] \leq \frac{\mathbb{E}\,|X|^p}{c^{p-1}} \leq \frac{M^p}{c^{p-1}} \to 0.$$

NB: The same proof works if S is $L^{1+\varepsilon}$-bounded for any $\varepsilon > 0$. □

4.5 Why Care About Uniform Integrability?

Q. Why care about UI?

A. UI + a.s. convergence $\implies$ L^1-convergence.

Theorem 4.7. UI and L^1-Convergence (TFAE)

Let $X_n \in L^1$ and $X_n \xrightarrow{\text{a.s.}} X$. Then the following are equivalent:

(i) $\{X_n : n \geq 1\}$ is UI.

(ii) $X \in L^1$ and $X_n \xrightarrow{L^1} X$.

(iii) $X \in L^1$ and $\mathbb{E}\,|X_n| \to \mathbb{E}\,|X|$.

Proof. **(i) $\implies$ (ii):** Define the truncation

$$X_n^c = X_n \mathbb{1}(|X_n| \leq c), \qquad X^c = X \mathbb{1}(|X| < c).$$

Note that $\mathbb{E}\,|X| \leq \liminf \mathbb{E}\,|X_n| < \infty$ (since UI implies L^1-bounded), so $X \in L^1$. We write

$$|X_n - X| \leq |X_n^c - X^c| + |X_n|\,\mathbb{1}(|X_n| \geq c) + |X|\,\mathbb{1}(|X| \geq c).$$

Taking expectations,

$$\mathbb{E}\,|X_n - X| \leq \underbrace{\mathbb{E}\,|X_n^c - X^c|}_{\to 0 \text{ by DCT}} + \underbrace{\mathbb{E}\Big[|X_n|\,\mathbb{1}(|X_n| \geq c)\Big]}_{\to 0 \text{ by UI}} + \underbrace{\mathbb{E}\Big[|X|\,\mathbb{1}(|X| \geq c)\Big]}_{\to 0 \text{ by DCT}}.$$

For the first term, $|X_n^c - X^c| \leq 2c$ and $X_n^c \to X^c$ a.s., so by the dominated convergence theorem, $\mathbb{E}\,|X_n^c - X^c| \to 0$ as $n \to \infty$.
For the second term, UI gives $\sup_n \mathbb{E}[|X_n|\,\mathbb{1}(|X_n| \geq c)] \to 0$ as $c \to \infty$.
For the third term, $\mathbb{E}[|X|\,\mathbb{1}(|X| \geq c)] \to 0$ as $c \to \infty$ by DCT.
So: send $n \to \infty$, then $c \to \infty$.

(ii) $\implies$ (iii): This is immediate since

$$\Big||X_n| - |X|\Big| \leq |X_n - X|,$$

so $\mathbb{E}\,|X_n| \to \mathbb{E}\,|X|$.

(iii) $\implies$ (i): We have

$$\mathbb{E}\Big[|X_n|\,\mathbb{1}(|X_n| > c)\Big] = \mathbb{E}\,|X_n| - \mathbb{E}\Big[|X_n|\,\mathbb{1}(|X_n| \leq c)\Big].$$

As $n \to \infty$:

- $\mathbb{E}\,|X_n| \to \mathbb{E}\,|X|$ by assumption.
- $\mathbb{E}[|X_n|\,\mathbb{1}(|X_n| \leq c)] \to \mathbb{E}[|X|\,\mathbb{1}(|X| \leq c)]$ (since $|X_n|\,\mathbb{1}(|X_n| \leq c) \leq c$ and converges a.s.; apply DCT — here we choose c to be a continuity point of X, i.e., $\mathbb{P}(|X| = c) = 0$).

Therefore,

$$\mathbb{E}\Big[|X_n|\,\mathbb{1}(|X_n| > c)\Big] \to \mathbb{E}\,|X| - \mathbb{E}\Big[|X|\,\mathbb{1}(|X| \leq c)\Big] = \mathbb{E}\Big[|X|\,\mathbb{1}(|X| > c)\Big].$$

Now let $c \to \infty$ to get the desired claim: $\mathbb{E}[|X|\,\mathbb{1}(|X| > c)] \to 0$, giving uniform integrability. $\square$

Next time: UI & Martingale convergence.

Chapter 5

L^1-Convergence (cont.), L^p-Convergence, Doob's Inequalities

Agenda

1. L^1-convergence of martingales.
2. L^p-convergence ($p > 1$).

Announcements

1. PSET 1 due Friday.
2. Survey.

5.1 Recall from Last Time

(i) **Uniformly integrable (UI) collections.** A collection $\{X_\alpha\}$ is UI if it is L^1-bounded and satisfies uniform absolute continuity: for every $\varepsilon > 0$ there exists $M > 0$ such that

$$\sup_\alpha \mathbb{E}\Big[|X_\alpha| \, \mathbb{1}(|X_\alpha| > M)\Big] < \varepsilon.$$

(ii) **Connection between a.s. convergence, L^1-convergence, and UI** (Durrett, Theorem 4.6.3):

If $X_n \xrightarrow{\text{a.s.}} X_\infty$, then $X_n \xrightarrow{L^1} X_\infty$ if and only if $\{X_n : n \geq 1\}$ is UI.

5.2 L^1-Convergence of Martingales

Theorem 5.1. L^1-Convergence of Martingales

Let $\{(X_n, \mathcal{F}_n) : n \geq 1\}$ be a martingale. Let $M = \sup_n \mathbb{E}\,|X_n|$ ($\leq \infty$). The following are equivalent:

(i) $M < \infty$ and $X_n \xrightarrow{\text{a.s.},L^1} X_\infty$.

(ii) $\mathbb{E}[X_\infty \mid \mathcal{F}_n] = X_n$ for all $n \geq 1$.

(iii) $M < \infty$ and $\mathbb{E}\,|X_\infty| = \lim_{n\to\infty} \mathbb{E}\,|X_n|$.

(iv) $\{X_n : n \geq 1\}$ is UI.

Proof. We prove a cycle of implications: (i) $\Rightarrow$ (ii) $\Rightarrow$ (iv) $\Rightarrow$ (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii): For all $A \in \mathcal{F}_n$, by the martingale property,

$$\mathbb{E}[X_{n+k}\mathbb{1}_A] = \mathbb{E}[X_n\mathbb{1}_A] \quad \forall\, k \geq 0.$$

As $k \to \infty$, we have $X_{n+k}\mathbb{1}_A \xrightarrow{\text{a.s.},L^1} X_\infty\mathbb{1}_A$ (since $X_{n+k} \xrightarrow{\text{a.s.},L^1} X_\infty$ by assumption). Therefore,

$$\lim_{k\to\infty} \mathbb{E}[X_{n+k}\mathbb{1}_A] = \mathbb{E}[X_\infty\mathbb{1}_A] = \mathbb{E}[X_n\mathbb{1}_A] \quad \forall\, A \in \mathcal{F}_n.$$

This shows $\mathbb{E}[X_\infty \mid \mathcal{F}_n] = X_n$.

(ii) $\Rightarrow$ (iv): By (ii), $X_n = \mathbb{E}[X_\infty \mid \mathcal{F}_n]$ for all $n \geq 1$. The collection $\{\mathbb{E}[X_\infty \mid \mathcal{F}_n] : n \geq 1\}$ is UI (proved last time).

(iv) $\Rightarrow$ (iii): Since $\{X_n\}$ is UI, in particular $M = \sup_n \mathbb{E}\,|X_n| < \infty$. By the Martingale Convergence Theorem (since $M < \infty$), $X_n \xrightarrow{\text{a.s.}} X_\infty$. Since $\{X_n\}$ is UI and $X_n \xrightarrow{\text{a.s.}} X_\infty$, by the connection between UI and L^1-convergence (Durrett, Theorem 4.6.3), $X_n \xrightarrow{L^1} X_\infty$. In particular, $\mathbb{E}\,|X_n| \to \mathbb{E}\,|X_\infty|$.

(iii) $\Rightarrow$ (i): Since $M < \infty$, the Martingale Convergence Theorem gives $X_n \xrightarrow{\text{a.s.}} X_\infty$. We have $\mathbb{E}\,|X_\infty| = \lim_n \mathbb{E}\,|X_n|$. By the theorem from last time (a.s. convergence plus convergence of norms implies L^1-convergence), $X_n \xrightarrow{L^1} X_\infty$. □

5.3 L^p-Convergence of Martingales ($p > 1$)

Theorem 5.2. L^p-Convergence of Martingales

Let $\{(X_n, \mathcal{F}_n) : n \geq 1\}$ be a martingale. Suppose

$$\sup_n \|X_n\|_p = \sup_n \Big(\mathbb{E}\,|X_n|^p\Big)^{1/p} < \infty \quad (:= M).$$

Then $X_n \xrightarrow{\text{a.s.},L^p} X_\infty$.

The proof relies on two lemmas.

Lemma 5.3. L^p-Boundedness Implies UI

If $S \subseteq L^1$ is L^p-bounded for some $p > 1$ (i.e., $\sup_{X\in S} \mathbb{E}\,|X|^p < \infty$), then S is uniformly integrable.

Lemma 5.4. Maximal Function in L^p

Let $\{(X_n, \mathcal{F}_n) : n \geq 1\}$ be an L^p-bounded martingale. Define $X^* = \sup_{n\geq 1} |X_n|$. Then $X^* \in L^p$.

5.3.1 Proof of the Theorem Assuming the Lemmas

Proof. By Lemma 5.3, L^p-boundedness implies $\{X_n\}$ is UI. By Theorem 5.1 (the L^1-convergence theorem, implication (iv) $\Rightarrow$ (i)), $X_n \xrightarrow{\text{a.s.},L^1} X_\infty$.
It remains to boost from L^1 to L^p. We have $|X_n - X_\infty|^p \to 0$ a.s. and $|X_n - X_\infty|^p \leq (2X^*)^p$, where $X^* = \sup_n |X_n| \in L^p$ by Lemma 5.4. By the Dominated Convergence Theorem,

$$\mathbb{E}\,|X_n - X_\infty|^p \to 0,$$

so $X_n \xrightarrow{L^p} X_\infty$. □

5.4 Doob's Inequalities

Theorem 5.5. Doob's Maximal Inequality

Let $\{(X_n, \mathcal{F}_n) : 0 \leq n \leq N\}$ be a submartingale (i.e., $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n$). Then for $\lambda > 0$,

$$\lambda\,\mathbb{P}\Big[\max_{0\leq j\leq N} X_j \geq \lambda\Big] \leq \mathbb{E}\Big[X_N \mathbb{1}\Big(\max_{0\leq j\leq N} X_j \geq \lambda\Big)\Big] \leq \mathbb{E}\,|X_N|\,.$$

Proof. Define the disjoint events

$$A_i = \{X_0 < \lambda,\, X_1 < \lambda,\, \ldots,\, X_{i-1} < \lambda,\, X_i \geq \lambda\}, \quad 0 \leq i \leq N.$$

Note $A_i \in \mathcal{F}_i$ and the A_i are disjoint. Moreover,

$$\Big\{\max_{0\leq j\leq N} X_j \geq \lambda\Big\} = \bigsqcup_{i=0}^{N} A_i.$$

On the event A_i, we have $X_i \geq \lambda$. By the submartingale property, $\mathbb{E}[X_N \mid \mathcal{F}_i] \geq X_i$, so

$$\mathbb{E}[X_N \mathbb{1}_{A_i}] \geq \mathbb{E}[X_i \mathbb{1}_{A_i}] \geq \lambda\,\mathbb{P}(A_i).$$

Summing over $i = 0, 1, \ldots, N$:

$$\mathbb{E}\Big[X_N \mathbb{1}\Big(\max_{0\leq j\leq N} X_j \geq \lambda\Big)\Big] = \sum_{i=0}^{N} \mathbb{E}[X_N \mathbb{1}_{A_i}] \geq \lambda \sum_{i=0}^{N} \mathbb{P}(A_i) = \lambda\,\mathbb{P}\Big[\max_{0\leq j\leq N} X_j \geq \lambda\Big].$$

This establishes the first inequality. The second inequality follows since $X_N \mathbb{1}(\cdot) \leq |X_N|$. □

Theorem 5.6. Doob's L^p-Inequality

Suppose $U, V \geq 0$ are random variables satisfying

$$\lambda\,\mathbb{P}[U \geq \lambda] \leq \mathbb{E}[V \mathbb{1}(U \geq \lambda)] \quad \forall\, \lambda > 0.$$

Then for $p > 1$,

$$\|U\|_p \leq \frac{p}{p-1}\,\|V\|_p\,.$$

Proof. We compute $\mathbb{E}[U^p]$ using the layer-cake representation and the hypothesis. We have

$$\mathbb{E}[U^p] = \int_0^\infty p\lambda^{p-1}\mathbb{P}[U \geq \lambda]\, d\lambda \leq \int_0^\infty p\lambda^{p-2}\mathbb{E}[V\mathbb{1}(U \geq \lambda)]\, d\lambda,$$

where we used $\lambda\mathbb{P}[U \geq \lambda] \leq \mathbb{E}[V\mathbb{1}(U \geq \lambda)]$, dividing both sides by λ and multiplying by $p\lambda^{p-1}$.
By Fubini's theorem,

$$\int_0^\infty p\lambda^{p-2}\mathbb{E}[V\mathbb{1}(U \geq \lambda)]\, d\lambda = \mathbb{E}\Big[V \int_0^U p\lambda^{p-2}\, d\lambda\Big] = \mathbb{E}\Big[V \cdot \frac{p}{p-1}U^{p-1}\Big] = \frac{p}{p-1}\mathbb{E}[VU^{p-1}].$$

Now apply Hölder's inequality with exponents p and $q = \frac{p}{p-1}$:

$$\mathbb{E}[VU^{p-1}] \leq \|V\|_p \cdot \left\|U^{p-1}\right\|_q = \|V\|_p \cdot \left(\mathbb{E}[U^{(p-1)q}]\right)^{1/q} = \|V\|_p \cdot \left(\mathbb{E}[U^p]\right)^{(p-1)/p}.$$

Therefore,

$$\mathbb{E}[U^p] \leq \frac{p}{p-1}\|V\|_p \cdot \left(\mathbb{E}[U^p]\right)^{(p-1)/p}.$$

We consider three cases:

(1) If $\mathbb{E}[U^p] = 0$, the result is trivial.

(2) If $0 < \mathbb{E}[U^p] < \infty$, divide both sides by $(\mathbb{E}[U^p])^{(p-1)/p}$:

$$\left(\mathbb{E}[U^p]\right)^{1/p} \leq \frac{p}{p-1}\|V\|_p,$$

which gives $\|U\|_p \leq \frac{p}{p-1}\|V\|_p$.

(3) If $\mathbb{E}[U^p] = \infty$, truncate: apply the argument to $U \wedge K$ (which is bounded, so case (2) applies), obtaining

$$\|U \wedge K\|_p \leq \frac{p}{p-1}\|V\|_p.$$

Send $K \to \infty$; by Monotone Convergence, $\|U\|_p \leq \frac{p}{p-1}\|V\|_p$. (If the right-hand side is infinite, the inequality holds trivially.)

□

5.4.1 Proof of Lemma 5.4

Proof. Let $\{(X_n, \mathcal{F}_n) : n \geq 1\}$ be an L^p-bounded martingale. Since $|X_n|$ is a submartingale (by Jensen's inequality), Doob's maximal inequality gives: for each N,

$$\lambda\,\mathbb{P}\Big[\max_{1\leq j\leq N}|X_j| \geq \lambda\Big] \leq \mathbb{E}\Big[|X_N|\,\mathbb{1}\Big(\max_{1\leq j\leq N}|X_j| \geq \lambda\Big)\Big].$$

Set $U_N = \max_{1\leq j\leq N}|X_j|$ and $V_N = |X_N|$. The hypothesis of Doob's L^p-inequality (Theorem 5.6) is satisfied, so

$$\|U_N\|_p \leq \frac{p}{p-1}\|X_N\|_p \leq \frac{p}{p-1}M.$$

As $N \to \infty$, $U_N \uparrow X^* = \sup_{n \geq 1} |X_n|$. By Monotone Convergence,

$$\|X^*\|_p = \lim_{N \to \infty} \|U_N\|_p \leq \frac{p}{p-1} M < \infty.$$

Hence $X^* \in L^p$. $\square$

Chapter 6

Reverse Martingales, Hewitt–Savage 0-1 Law, U-Statistics

Agenda

1. Doob maximal inequality.
2. Reverse martingales.
3. Hewitt–Savage 0/1 law.

Announcements

1. HW1 due Sunday.
2. Survey goes out.

6.1 Reverse Martingales

Definition 6.1. Reverse Martingale

Let $\{(X_n, \mathcal{F}_n) : n \leq 0\}$ be an adapted sequence with $X_n \in L^1$ for all $n \leq 0$. The filtration satisfies

$$\mathcal{F}_{-\infty} = \bigcap_{n \leq 0} \mathcal{F}_n \subseteq \cdots \subseteq \mathcal{F}_{-1} \subseteq \mathcal{F}_0.$$

We say $\{(X_n, \mathcal{F}_n) : n \leq 0\}$ is a *reverse martingale* (RMG) if

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n \quad \forall\, n \leq -1.$$

Theorem 6.2. Characterization and Convergence of Reverse Martingales

Let $X_0 \in L^1$. Then $\{(X_n, \mathcal{F}_n) : n \leq 0\}$ is a reverse martingale if and only if

$$X_n = \mathbb{E}[X_0 \mid \mathcal{F}_n] \quad \forall\, n \leq 0.$$

Further,

$$\mathbb{E}[X_0 \mid \mathcal{F}_n] \xrightarrow{\text{a.s.}} \mathbb{E}[X_0 \mid \mathcal{F}_{-\infty}].$$

Proof. ($\Rightarrow$) We prove by induction on $-n$.
Base case ($n = -1$): By definition,

$$X_{-1} = \mathbb{E}[X_0 \mid \mathcal{F}_{-1}].$$

Inductive step: Suppose $X_{n+1} = \mathbb{E}[X_0 \mid \mathcal{F}_{n+1}]$. Then

$$X_n = \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}\Big[\mathbb{E}[X_0 \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\Big] = \mathbb{E}[X_0 \mid \mathcal{F}_n],$$

where the last equality follows by the tower property (since $\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$).
($\Leftarrow$) Suppose $X_n = \mathbb{E}[X_0 \mid \mathcal{F}_n]$ for all $n \leq 0$. Then for $n \leq -1$,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}\Big[\mathbb{E}[X_0 \mid \mathcal{F}_{n+1}] \mid \mathcal{F}_n\Big] = \mathbb{E}[X_0 \mid \mathcal{F}_n] = X_n,$$

again by the tower property. □

6.2 Doob's Upcrossing Inequality for Reverse Martingales

For the reverse martingale $\{(X_n, \mathcal{F}_n), \ldots, (X_0, \mathcal{F}_0)\}$, Doob's upcrossing inequality gives

$$\mathbb{E}\Big[U_n([a,b])\Big] \leq \frac{1}{b-a}\,\mathbb{E}\Big[(X_0 - a)_+\Big].$$

By the monotone convergence theorem (MCT),

$$\mathbb{E}\Big[U_\infty([a,b])\Big] \leq \frac{\mathbb{E}\Big[(X_0 - a)_+\Big]}{b-a}.$$

Almost sure convergence follows by the same argument as in the Martingale Convergence Theorem.

6.3 Boosting to L^1-Convergence

The family $\{\mathbb{E}[X_0 \mid \mathcal{F}_n] : n \leq 0\}$ is uniformly integrable (UI). Therefore,

$$X_n \xrightarrow{\text{a.s.}} X_{-\infty} \quad \text{and} \quad X_n \xrightarrow{L^1} X_{-\infty}.$$

6.4 Showing $X_{-\infty} = \mathbb{E}[X_0 \mid \mathcal{F}_{-\infty}]$

We verify the two conditions:
Observation 1. $X_{-\infty}$ is $\mathcal{F}_{-\infty}$-measurable.
Since X_n is $\mathcal{F}_n$-measurable for each $n \leq 0$, and $X_n \xrightarrow{\text{a.s.}} X_{-\infty}$, the limit $X_{-\infty}$ is measurable with respect to $\bigcap_{n \leq 0} \mathcal{F}_n = \mathcal{F}_{-\infty}$.
Observation 2. We use L^1-convergence and the π-λ theorem.
For any $A \in \mathcal{F}_n$ (for any fixed $n \leq 0$),

$$\mathbb{E}[X_0 \mathbb{1}_A] = \mathbb{E}\Big[\mathbb{E}[X_0 \mid \mathcal{F}_n]\, \mathbb{1}_A\Big] = \mathbb{E}[X_n \mathbb{1}_A].$$

By L^1-convergence, $\mathbb{E}[X_n \mathbb{1}_A] \to \mathbb{E}[X_{-\infty} \mathbb{1}_A]$. Hence

$$\mathbb{E}[X_0 \mathbb{1}_A] = \mathbb{E}[X_{-\infty} \mathbb{1}_A] \quad \forall\, A \in \bigcup_{n \le 0} \mathcal{F}_n.$$

By the π-λ theorem, this extends to all $A \in \mathcal{F}_{-\infty}$, so $X_{-\infty} = \mathbb{E}[X_0 \mid \mathcal{F}_{-\infty}]$.

6.5 Application: Convergence of U-Statistics of i.i.d. Random Variables

Let $\{X_n : n \ge 1\}$ be an i.i.d. sequence and let $\varphi \colon \mathbb{R}^\ell \to \mathbb{R}$ be bounded and measurable.

Definition 6.3. U-Statistic

The *U-statistic* of order ℓ based on a sample of size m is

$$\hat{S}_m(\varphi) = \frac{1}{(m)_\ell} \sum_{\substack{i_1,\dots,i_\ell \in \{1,\dots,m\} \\ \text{all distinct}}} \varphi(X_{i_1}, \dots, X_{i_\ell}),$$

where $(m)_\ell = m(m-1)\cdots(m-\ell+1)$ is the falling factorial.

Theorem 6.4. Convergence of U-Statistics

If $\{X_n : n \ge 1\}$ are i.i.d., then

$$\hat{S}_m(\varphi) \xrightarrow{\text{a.s.}} \mathbb{E}[\varphi(X_1, \dots, X_\ell)] \quad \text{and} \quad \hat{S}_m(\varphi) \xrightarrow{L^1} \mathbb{E}[\varphi(X_1, \dots, X_\ell)].$$

Proof. **Step 1: Construct a reverse filtration.**
For each $m \ge 1$, define

$$\mathcal{F}_{-m} = \sigma\left(\frac{1}{(m)_\ell} \sum_{\substack{i_1,\dots,i_\ell \in \{1,\dots,m\} \\ \text{all distinct}}} \tilde{\varphi}(X_{i_1}, \dots, X_{i_\ell}) : \tilde{\varphi} \colon \mathbb{R}^\ell \to \mathbb{R} \text{ bounded measurable} \right).$$

Intuition: $\mathcal{F}_{-m}$ captures the information in $\{X_1, \dots, X_m\}$ up to a random permutation.

If $n \ge m$, then $\mathcal{F}_{-n} \subseteq \mathcal{F}_{-m}$, so

$$\mathcal{F}_{-1} \supseteq \mathcal{F}_{-2} \supseteq \cdots.$$

Let $\mathcal{E} = \bigcap_{n \ge 1} \mathcal{F}_{-n}$ denote the *exchangeable σ-algebra.*

Step 2: Show $\hat{S}_m(\varphi) = \mathbb{E}[\varphi(X_1, \dots, X_\ell) \mid \mathcal{F}_{-m}]$.
By symmetry of the i.i.d. sequence and the definition of $\mathcal{F}_{-m}$,

$$\hat{S}_m(\varphi) = \mathbb{E}\big[\hat{S}_m(\varphi) \mid \mathcal{F}_{-m}\big] = \mathbb{E}\big[\varphi(X_1, \dots, X_\ell) \mid \mathcal{F}_{-m}\big].$$

Step 3: Apply the reverse martingale convergence theorem.
By the reverse martingale convergence theorem,

$$\hat{S}_m(\varphi) \xrightarrow{\text{a.s.}} \mathbb{E}[\varphi(X_1, \dots, X_\ell) \mid \mathcal{E}] \quad \text{and} \quad \hat{S}_m(\varphi) \xrightarrow{L^1} \mathbb{E}[\varphi(X_1, \dots, X_\ell) \mid \mathcal{E}].$$

Step 4: Show $\mathbb{E}[\varphi(X_1, \ldots, X_\ell) \mid \mathcal{E}] = \mathbb{E}[\varphi(X_1, \ldots, X_\ell)]$.
For each $n \geq 1$, define

$$\hat{S}_{m,n}(\varphi) = \frac{1}{(m)_\ell} \sum_{\substack{i_1,\ldots,i_\ell \in \{n+1,\ldots,m\} \\ \text{all distinct}}} \varphi(X_{i_1}, \ldots, X_{i_\ell}),$$

where all indices are restricted to be $> n$. Then

$$\left|\hat{S}_m(\varphi) - \hat{S}_{m,n}(\varphi)\right| \leq \frac{C(n)}{m} \|\varphi\|_\infty \to 0 \quad \text{as } m \to \infty,$$

where $C(n)$ depends only on n and ℓ.
Letting $m \to \infty$, we obtain $\hat{S}_\infty(\varphi) \perp\!\!\!\perp \{X_1, \ldots, X_n\}$ for all $n \geq 1$.

Therefore,

$$\mathrm{Var}\Big(\mathbb{E}[\varphi(X_1, \ldots, X_\ell) \mid \mathcal{E}]\Big) = 0,$$

which implies $\mathbb{E}[\varphi(X_1, \ldots, X_\ell) \mid \mathcal{E}] = \mathbb{E}[\varphi(X_1, \ldots, X_\ell)]$ a.s. $\square$

Chapter 7

SLLN with Reverse Martingales, Brownian Motion

Agenda

1. SLLN with reverse martingales.
2. Brownian motion.

Announcements. Survey on Ed. Midterm March 9.

7.1 Brownian Motion

7.1.1 Historical Background

- **Brown (1828):** Observed erratic motion of pollen grains suspended in water.
- **Bachelier (1900):** First mathematical model of Brownian motion, motivated by finance (modeling stock prices).
- **Einstein (1905):** Independently developed a theory of Brownian motion from the perspective of physics.
- **Wiener (1923):** Gave the first rigorous mathematical construction.

7.1.2 Definition of Standard Brownian Motion

Definition 7.1. Standard Brownian Motion

A stochastic process $\{B_t : t \in [0,1]\}$ (where for each t, $B_t \colon (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$) is a *standard Brownian motion* (sBM) if:

(i) $B_0 = 0$, $B_t \sim N(0,t)$ for all $t \in [0,1]$, and $B_t - B_s \sim N(0, t-s)$ for all $0 \le s \le t \le 1$.

(ii) *Independent increments:* For all $0 \le t_1 \le t_2 \le \cdots \le t_n \le 1$, the random variables

$$B_{t_1},\ B_{t_2} - B_{t_1},\ \ldots,\ B_{t_n} - B_{t_{n-1}}$$

are independent.

(iii) *Continuous paths:* The map $t \mapsto B_t(\omega)$ is continuous for $\mathbb{P}$-almost every ω.

Remark 7.2

Recall that $(C([0,1]), d_{\sup})$ is a complete, separable metric space, where $d_{\sup}(f,g) = \sup_{t\in[0,1]} |f(t) - g(t)|$.

7.1.3 Equivalent Definitions

Definition 7.3. Equivalent Definition 1: $C([0,1])$-Valued Random Variable

A standard Brownian motion is a $C([0,1])$-valued random variable satisfying conditions (i) and (ii) of Definition 7.1.

Definition 7.4. Equivalent Definition 2: Law of Brownian Motion (Wiener Measure)

The *law of Brownian motion* (or *Wiener measure*) is a probability measure μ on $(C([0,1]), \mathcal{B}_{C([0,1])})$ such that the coordinate process $X_t(\omega) = \omega(t)$ satisfies conditions (i) and (ii) of Definition 7.1.

7.1.4 Existence and Uniqueness

Q. Does standard Brownian motion exist? Is it unique (in law)?

7.2 Uniqueness of Brownian Motion

Theorem 7.5. Uniqueness of Brownian Motion

Let μ and ν be probability measures on $(C([0,1]), \mathcal{B}_{C([0,1])})$ both satisfying conditions (i) and (ii) of Definition 7.1. Then

$$\mu(A) = \nu(A) \quad \text{for all } A \in \mathcal{B}_{C([0,1])}.$$

Proof. The strategy is to approximate with finite-dimensional distributions and use continuity.
Construct approximating sequences $\{B_1^{(k)} : k \geq 1\}$ and $\{B_2^{(k)} : k \geq 1\}$ (piecewise linear interpolations on dyadic grids) such that:

(a) $\mathcal{L}(B_1^{(k)}) = \mathcal{L}(B_2^{(k)})$ for all $k \geq 1$ (since the finite-dimensional distributions are determined by (i) and (ii)).

(b) $d_{\sup}(B_i^{(k)}, B_i) \to 0$ almost surely as $k \to \infty$, for $i = 1, 2$.

Since $B_1^{(k)} \overset{d}{=} B_2^{(k)}$ for all k, and $B_i^{(k)} \to B_i$ almost surely in $C([0,1])$, it follows that $B_1 \overset{d}{=} B_2$.
Indeed, for any bounded continuous function $f\colon C([0,1]) \to \mathbb{R}$:

$$\mathbb{E}[f(B_1)] = \lim_{k\to\infty} \mathbb{E}[f(B_1^{(k)})] = \lim_{k\to\infty} \mathbb{E}[f(B_2^{(k)})] = \mathbb{E}[f(B_2)].$$

Hence $B_1 \stackrel{d}{=} B_2$, i.e., $\mu = \nu$. (Why does the first and last equality hold? By the dominated convergence theorem, since f is bounded and $f(B_i^{(k)}) \to f(B_i)$ a.s. by continuity of f.) $\square$

Chapter 8

Existence of Brownian Motion, Scaling Properties

Agenda

1. Existence of Brownian motion.
2. Brownian motion on $[0, \infty)$.

Announcements. Problem set 2 online. Midterm March 9.

8.1 Recap

Last time we defined standard Brownian motion (sBM) on $[0, 1]$ and proved *uniqueness* of the Wiener measure on $C([0, 1])$.

Today: Existence. Equivalently, can we construct a measurable map $B\colon (\Omega, \mathcal{F}) \to (C([0, 1]), \mathcal{B}_{C([0,1])})$ satisfying the sBM properties?

Remark 8.1. Gaussian Process Characterization

Standard Brownian motion on $[0, 1]$ is equivalently a Gaussian process with $\mathbb{E}[B_t] = 0$, $\mathrm{Cov}(B_t, B_s) = s \wedge t$, and continuous paths almost surely.

8.2 Existence of Brownian Motion (Paul Lévy's Construction)

Theorem 8.2. Existence of Brownian Motion

Standard Brownian motion on $[0, 1]$ exists.

Proof. Start with an arbitrary probability space supporting iid standard normal random variables $Z_1, Z_2, \ldots$ (i.e., $Z_i \sim N(0, 1)$ iid).
Strategy: Construct a sequence $\{B^{(1)}, B^{(2)}, \ldots\} \subset C([0, 1])$ using $\{Z_n\}$, and show this sequence converges to a limit $B \in C([0, 1])$ satisfying all sBM properties.

Step 1. Define $B^{(1)} \in C([0,1])$ by:

$$B_t^{(1)} = \begin{cases} 0 & \text{if } t = 0, \\ Z_1 & \text{if } t = 1, \end{cases}$$

with linear interpolation between 0 and 1. That is, $B_t^{(1)} = tZ_1$.

Step 2. Define $B_{1/2}$. From the sBM properties, given $B_0 = 0$ and $B_1 = Z_1$, we need

$$B_{1/2} = \frac{1}{2}Z_1 + \widetilde{Z}, \qquad \widetilde{Z} \sim N\Big(0, \frac{1}{4}\Big) \text{ independent of } Z_1.$$

Define $B^{(2)} \in C([0,1])$ by:

$$B_t^{(2)} = \begin{cases} 0 & \text{if } t = 0, \\ B_{1/2} & \text{if } t = 1/2, \\ Z_1 & \text{if } t = 1, \end{cases}$$

with linear interpolation between consecutive points.
Check: $(B_0^{(2)}, B_{1/2}^{(2)}, B_1^{(2)}) = (0, B_{1/2}, Z_1)$ has the correct joint distribution for sBM evaluated at $\{0, 1/2, 1\}$.

General construction (Step $k+1$). Suppose we have already defined the values $B_0, B_{1/2^k}, B_{2/2^k}, \ldots, B_1$ at all dyadic rationals of order k. For each *odd* integer j with $1 \le j \le 2^{k+1} - 1$, define

$$B_{j/2^{k+1}} = \frac{1}{2}\Big(B_{(j-1)/2^{k+1}} + B_{(j+1)/2^{k+1}}\Big) + Z_{j/2^{k+1}},$$

where

$$Z_{j/2^{k+1}} \sim N\Big(0, \frac{1}{2^{k+2}}\Big),$$

all independent of each other and of everything previously defined. Then $B^{(k+1)}$ is the piecewise linear interpolation through all dyadic points of order $k+1$.

Convergence. We claim that $\{B^{(k)}\}$ converges in $C([0,1])$.

> **Claim:** $\mathbb{E}\Big[\sum_{k=1}^{\infty} \big\|B^{(k+1)} - B^{(k)}\big\|_\infty\Big] < \infty$, which implies $\{B^{(k)}\}$ is almost surely Cauchy in $(C([0,1]), d_{\sup})$.

Proof of claim. By construction, $B^{(k+1)}$ and $B^{(k)}$ agree at all dyadic rationals of order k, and between consecutive such points, $B^{(k+1)}$ deviates from $B^{(k)}$ by at most the maximum of the added Gaussian increments. Hence

$$\big\|B^{(k+1)} - B^{(k)}\big\|_\infty \le \max_{j \text{ odd}} \big|Z_{j/2^{k+1}}\big|,$$

where the maximum is over at most 2^k independent centered Gaussians, each with variance $1/2^{k+2}$.
Using the standard bound on the expected maximum of Gaussians:

$$\mathbb{E}\Big[\max_{j \text{ odd}} \big|Z_{j/2^{k+1}}\big|\Big] \le \frac{C\sqrt{k}}{2^{k/2}}$$

for a universal constant $C > 0$. Since $\sum_{k=1}^{\infty} \frac{C\sqrt{k}}{2^{k/2}} < \infty$, the claim follows.

Since $(C([0,1]), d_{\sup})$ is a complete metric space and $\{B^{(k)}\}$ is almost surely Cauchy, the limit

$$B = \lim_{k\to\infty} B^{(k)}$$

exists in $C([0,1])$ almost surely.

Verification. The limit B satisfies all sBM properties:

- *Continuous paths:* $B \in C([0,1])$ by completeness.
- *Gaussian increments:* At each stage k, $B^{(k)}$ has the correct finite-dimensional distributions on dyadic rationals of order k (by induction). By continuity, B has the correct finite-dimensional distributions at all times.
- *Independent increments:* Follows from the finite-dimensional distributions.

This completes the proof of existence. □

8.3 Brownian Motion on $[0, \infty)$

Definition 8.3. Standard Brownian Motion on $[0, \infty)$

A stochastic process $\{B_t : t \geq 0\}$ is a *standard Brownian motion on* $[0, \infty)$ if:

(i) $B_0 = 0$ and $B_t - B_s \sim N(0, t-s)$ for all $0 \leq s \leq t < \infty$.

(ii) *Independent increments:* For all $n \geq 1$ and all $0 \leq t_1 \leq t_2 \leq \cdots \leq t_n$,

$$(B_{t_1},\ B_{t_2} - B_{t_1},\ \ldots,\ B_{t_n} - B_{t_{n-1}})$$

are independent.

(iii) *Continuous paths:* $t \mapsto B_t$ is continuous almost surely.

8.3.1 Construction and Uniqueness

Construction: "Glue" together iid standard Brownian motions on $[0,1]$.

Let $B^1, B^2, B^3, \ldots$ be iid sBMs on $[0,1]$. For $k \leq t \leq k+1$ (where $k \in \{0,1,2,\ldots\}$), define

$$B_t = B^{k+1}_{t-k} + B^k_1 + B^{k-1}_1 + \cdots + B^1_1 = B^{k+1}_{t-k} + \sum_{i=1}^{k} B^i_1.$$

Check: $\{B_t : t \geq 0\}$ satisfies all three properties of sBM on $[0, \infty)$.
Uniqueness: The same argument as in the $[0,1]$ case extends to $[0, \infty)$.

Remark 8.4. $C([0, \infty))$ as a Metric Space

Let $C([0,\infty)) = \{f\colon [0,\infty) \to \mathbb{R} : f \text{ continuous}\}$. Define

$$d(f,g) = \sum_{n=0}^{\infty} \frac{1}{2^n} \cdot \frac{\|f-g\|_{[n,n+1]}}{1 + \|f-g\|_{[n,n+1]}},$$

where $\|f-g\|_{[n,n+1]} = \sup_{t\in[n,n+1]} |f(t)-g(t)|$. This metric induces uniform convergence on compact sets, and $(C([0,\infty)), d)$ is a complete, separable metric space. Moreover, $\{B_t : t \geq 0\}$ is a $C([0,\infty))$-valued random variable.

8.4 Scaling Properties of Brownian Motion

Theorem 8.5. Scaling Properties of Brownian Motion

Let $\{B_t : t \geq 0\} \sim$ sBM. Then:

(i) *Diffusive scaling:* For any $a > 0$, the process $X_t = \frac{1}{a} B_{a^2 t}$ is also a standard Brownian motion.

(ii) *Stationarity of increments:* For any $s \geq 0$, the process $\{B_{t+s} - B_s : t \geq 0\}$ is also a standard Brownian motion.

(iii) *Time inversion:* Define

$$X_t = \begin{cases} 0 & \text{if } t = 0, \\ t\, B_{1/t} & \text{if } t > 0. \end{cases}$$

Then $\{X_t : t \geq 0\} \sim$ sBM.

Proof. We prove part (iii); parts (i) and (ii) follow from direct verification of the sBM properties.

Proof of (iii).

(a) $X_0 = 0$ by definition.

(b) *Finite-dimensional distributions:* We show that $(X_{t_1}, \ldots, X_{t_k})$ is multivariate Gaussian with the correct covariance structure.

Since $X_t = tB_{1/t}$ for $t > 0$, and B is a Gaussian process, $(X_{t_1}, \ldots, X_{t_k})$ is multivariate Gaussian (as a linear function of the Gaussian vector $(B_{1/t_1}, \ldots, B_{1/t_k})$).

We compute the covariance: for $s, t > 0$,

$$\mathbb{E}[X_t X_s] = st \cdot \mathbb{E}[B_{1/t} B_{1/s}] = st \cdot \min\left\{\frac{1}{s}, \frac{1}{t}\right\} = st \cdot \frac{1}{\max\{s,t\}} = \min\{s,t\}.$$

This is exactly the covariance structure of sBM, and $\mathbb{E}[X_t] = t \cdot \mathbb{E}[B_{1/t}] = 0$.

(c) *Continuity of paths:*

For $t > 0$: $X_t = tB_{1/t}$ is continuous by construction (since $t \mapsto 1/t$ is continuous on $(0,\infty)$ and B has continuous paths).

At $t = 0$: We need to show $X_t \to 0$ as $t \downarrow 0$ almost surely. Note that $X_t = tB_{1/t}$, so we need $tB_{1/t} \to 0$.

Along rationals: $\{X_t : t \in (0,\infty) \cap \mathbb{Q}\} \overset{d}{=} \{B_t : t \in (0,\infty) \cap \mathbb{Q}\}$, so $X_t \to 0$ along rationals almost surely (since $B_t \to B_0 = 0$ as $t \downarrow 0$).

Combining with the fact that $t \mapsto X_t$ is continuous on $(0, \infty)$, we conclude $X_t \to 0$ as $t \downarrow 0$ almost surely. $\square$

Chapter 9

Properties of Brownian Motion

9.1 Almost Sure Nowhere Differentiability of Brownian Motion

Recall the upper and lower right derivatives.

Definition 9.1. Upper and Lower Right Derivatives

Let $f\colon [0,\infty) \to \mathbb{R}$. Define the *upper right derivative* and *lower right derivative* of f at t by

$$D^* f(t) = \limsup_{h\downarrow 0} \frac{f(t+h)-f(t)}{h}, \qquad D_* f(t) = \liminf_{h\downarrow 0} \frac{f(t+h)-f(t)}{h}.$$

Theorem 9.2. Nowhere Differentiability of Brownian Motion (Paley, Wiener, Zygmund, 1933)

Brownian motion is nowhere differentiable almost surely. More precisely, almost surely, for all $t \geq 0$, either $D^*B(t) = +\infty$ or $D_*B(t) = -\infty$ (or both).

Proof. Suppose for the sake of contradiction that there exists $t_0 \in [0,1]$ such that

$$-\infty < D_*B(t_0) \leq D^*B(t_0) < \infty.$$

Then

$$\limsup_{h\downarrow 0} \frac{|B(t_0+h)-B(t_0)|}{h} < \infty$$

and $\{B_t : t \in [0,2]\}$ is bounded (by continuity of paths on a compact set).
This implies there exists $M < \infty$ such that

$$\sup_{h\in[0,1]} \frac{|B(t_0+h)-B(t_0)|}{h} \leq M.$$

We will show this event has probability 0.

Fix $M > 0$**.** For $n \geq 1$, consider the dyadic partition $\{k/2^n : k = 0,1,\ldots,2^n\}$ of $[0,1]$. If $t_0 \in [k/2^n, (k+1)/2^n)$ for some k, then for $j = 1,2,3$:

$$\left|B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right)\right| \leq \left|B\left(\frac{k+j}{2^n}\right) - B(t_0)\right| + \left|B\left(\frac{k+j-1}{2^n}\right) - B(t_0)\right|.$$

Since $|B(t_0+h)-B(t_0)| \le Mh$ for $h \in [0,1]$, and $0 \le \frac{k+j}{2^n} - t_0 \le \frac{j+1}{2^n}$ and $0 \le \frac{k+j-1}{2^n} - t_0 \le \frac{j}{2^n}$, we get:

- For $j=1$:
$$\left|B\left(\frac{k+1}{2^n}\right) - B\left(\frac{k}{2^n}\right)\right| \le \frac{3M}{2^n}.$$
- For $j>1$:
$$\left|B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right)\right| \le \frac{M(2j+1)}{2^n}.$$

Define the event

$$\Omega_{n,k} = \left\{\left|B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right)\right| \le \frac{M(2j+1)}{2^n} \text{ for } j=1,2,3\right\}.$$

Bounding $\mathbb{P}(\Omega_{n,k})$: By independence of the increments,

$$\mathbb{P}(\Omega_{n,k}) \le \prod_{j=1}^{3} \mathbb{P}\left[\left|B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right)\right| \le \frac{M(2j+1)}{2^n}\right].$$

Since $B\left(\frac{k+j}{2^n}\right) - B\left(\frac{k+j-1}{2^n}\right) \sim N(0,1/2^n)$, we write this increment as $\frac{1}{2^{n/2}}Z$ where $Z \sim N(0,1)$. Then

$$\mathbb{P}\left[\left|\frac{1}{2^{n/2}}Z\right| \le \frac{M(2j+1)}{2^n}\right] = \mathbb{P}\left[|Z| \le \frac{M(2j+1)}{2^{n/2}}\right] \le \frac{M(2j+1)}{2^{n/2}} \cdot \sqrt{\frac{2}{\pi}} \le \frac{M(2j+1)}{2^{n/2}}.$$

Using $2j+1 \le 7$ for $j=1,2,3$:

$$\mathbb{P}(\Omega_{n,k}) \le \left(\frac{7M}{2^{n/2}}\right)^3.$$

Union bound: Summing over $k=0,1,\dots,2^n-1$:

$$\mathbb{P}\left[\bigcup_{k=0}^{2^n-1} \Omega_{n,k}\right] \le 2^n \cdot \left(\frac{7M}{2^{n/2}}\right)^3 = (7M)^3 \cdot 2^n \cdot 2^{-3n/2} = \frac{(7M)^3}{2^{n/2}}.$$

This is summable in n.

Applying Borel–Cantelli: Since $\sum_{n=1}^{\infty} \frac{(7M)^3}{2^{n/2}} < \infty$, by the first Borel–Cantelli lemma,

$$\mathbb{P}\left[\exists\, t_0 \in [0,1] \text{ with } \sup_{h\in[0,1]} \frac{|B(t_0+h)-B(t_0)|}{h} \le M\right] \le \mathbb{P}\left[\bigcup_k \Omega_{n,k} \text{ for infinitely many } n\right] = 0.$$

Finally, let $M \uparrow \infty$ (through integers):

$$\mathbb{P}[\exists\, t_0 \in [0,1] \text{ with } D^*B(t_0) < \infty \text{ and } D_*B(t_0) > -\infty] \le \sum_{M=1}^{\infty} 0 = 0. \qquad \square$$

9.2 Markov Property of Brownian Motion

Recall: $\{X_n : n \geq 1\}$ is a Markov chain if

$$\mathbb{P}[X_{n+1} \in A \mid X_0, \ldots, X_n] = \mathbb{P}[X_{n+1} \in A \mid X_n].$$

Recasting: Equivalently,

$$\mathbb{P}[X_{n+1} = x_{n+1},\, X_{n-k} = x_{n-k} \mid X_n = x_n] = \mathbb{P}[X_{n+1} = x_{n+1} \mid X_n = x_n] \cdot \mathbb{P}[X_{n-k} = x_{n-k} \mid X_n = x_n].$$

Moral: Given the present, the future and the past are independent.

9.3 Filtrations

Definition 9.3. Filtration

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A *filtration* $\{\mathcal{F}_t : t \geq 0\}$ is a collection of σ-algebras such that

$$\mathcal{F}_s \subseteq \mathcal{F}_t \quad \text{for all } s \leq t.$$

Example 9.4. Natural Filtration of Brownian Motion

Define $\mathcal{F}_t^0 = \sigma(\{B_s : s \leq t\})$.

Lemma 9.5. Shifted Brownian Motion

Let $\{B_t : t \geq 0\} \sim$ sBM and define $W_t = B_{t+s} - B_s$. Then $\{W_t : t \geq 0\}$ is a standard Brownian motion independent of $\{B_t : 0 \leq t \leq s\}$.
Formally: for all $A \in \mathcal{B}(C([0,s]))$ measurable and $B \in \mathcal{B}(C([0,\infty)))$ measurable,

$$\mathbb{P}[(W_t)_{t\geq 0} \in B,\ (B_t)_{t \leq s} \in A] = \mathbb{P}[(B_t)_{t\leq s} \in A] \cdot \mathbb{P}[(W_t)_{t\geq 0} \in B]\,.$$

Proof. Let $\{X_t : 0 \leq t \leq s\}$ and $\{Y_t : t \geq 0\}$ be independent Brownian motions. Define

$$B_t^* = \begin{cases} X_t & \text{if } 0 \leq t \leq s, \\ X_s + Y_{t-s} & \text{if } t > s. \end{cases}$$

Then $\{B_t^* : t \geq 0\}$ is a standard Brownian motion (by the gluing construction).
Note that $(B_t^*)_{t\leq s} = (X_t)_{t \leq s}$ and $B_{t+s}^* - B_s^* = Y_t$, so the shifted process equals $\{Y_t : t \geq 0\}$, which is independent of $(X_t)_{t\leq s}$ by construction.
By independence of X and Y, for measurable A and B:

$$\mathbb{P}[(Y_t)_{t\geq 0} \in B,\ (X_t)_{t\leq s} \in A] = \mathbb{P}[(X_t)_{t\leq s} \in A] \cdot \mathbb{P}[(Y_t)_{t\geq 0} \in B]\,.$$

Since $\{B_t^* : t \geq 0\} \overset{d}{=} \{B_t : t \geq 0\}$, the same factorization holds for (B_t). It remains to note that $\sigma(\{f : f(t_i) \in A_i, i = 1, \ldots, k\}) = \mathcal{B}(C([0,s]))$ (by a π-λ argument), completing the proof. $\square$

Chapter 10

Markov Property, Stopping Times, Strong Markov Property

Agenda

1. Markov property of BM.
2. Stopping times.
3. Strong Markov property.

Announcements. Midterm March 9. Practice problems (solutions to be posted).

10.1 Right-Continuous Filtrations

Definition 10.1. Right-Continuous Filtration

A filtration $\{\mathcal{F}_t^+ : t \geq 0\}$ is *right-continuous* if

$$\mathcal{F}_t^+ = \bigcap_{s>t} \mathcal{F}_s^+.$$

Example 10.2. Canonical Right-Continuous Filtration

Let $\mathcal{F}_t^0 = \sigma(\{B_s : s \leq t\})$. Define

$$\mathcal{F}_t^+ = \bigcap_{s>t} \mathcal{F}_s^0.$$

Check: $\{\mathcal{F}_t^+ : t \geq 0\}$ is right-continuous.

Why care about right-continuous filtrations?
A: Plays well with the notion of stopping times, and does not change the picture too much (BM is Markov with respect to $\mathcal{F}_t^+$, and Blumenthal's 0/1 law holds).

10.2 Markov Property with $\mathcal{F}_t^+$

First let us check that the Markov property continues to hold with $\{\mathcal{F}_t^+ : t \geq 0\}$.

Lemma 10.3. Shifted BM Independent of $\mathcal{F}_s^+$

Let $\{B_t : t \geq 0\} \sim$ sBM. Define $W_t = B_{t+s} - B_s$ for $t \geq 0$. Then $\{W_t : t \geq 0\}$ is a standard Brownian motion and $\{W_t\} \perp\!\!\!\perp \mathcal{F}_s^+$.

Proof. For $n \geq 1$, define $W_t^n = B_{t+s+\frac{1}{n}} - B_{s+\frac{1}{n}}$. Then $W_t^n \perp\!\!\!\perp \mathcal{F}^0_{s+\frac{1}{n}}$.
Since $\mathcal{F}_s^+ \subseteq \mathcal{F}^0_{s+\frac{1}{n}}$, we have $\{W_t^n : t \geq 0\} \perp\!\!\!\perp \mathcal{F}_s^+$ for all $n \geq 1$.
By continuity, as $n \to \infty$, $W_t^n \to W_t$. Therefore $\{W_t : t \geq 0\} \perp\!\!\!\perp \mathcal{F}_s^+$. □

10.3 Stopping Times

Next, let us look at the relation with stopping times.

Definition 10.4. Stopping Time

A non-negative random variable T, defined on the same probability space as B, is called a *stopping time* for $\{(B_t, \mathcal{F}_t^+) : t \geq 0\}$ if

$$\{T < t\} \in \mathcal{F}_t^+ \quad \forall\, t \geq 0.$$

Claim 10.5. Equivalent Characterization of Stopping Times

The definition $\{T < t\} \in \mathcal{F}_t^+$ for all $t \geq 0$ is equivalent to $\{T \leq t\} \in \mathcal{F}_t^+$ for all $t \geq 0$.

Proof. ($\Leftarrow$) Suppose $\{T \leq t\} \in \mathcal{F}_t^+$ for all $t \geq 0$. Then

$$\{T < t\} = \bigcup_{n \geq 1} \left\{T \leq t - \tfrac{1}{n}\right\}.$$

Since $\mathcal{F}^+_{t-\frac{1}{n}} \subseteq \mathcal{F}_t^+$, each $\{T \leq t - \frac{1}{n}\} \in \mathcal{F}_t^+$, so $\{T < t\} \in \mathcal{F}_t^+$.

($\Rightarrow$) Suppose $\{T < t\} \in \mathcal{F}_t^+$ for all $t \geq 0$. Then

$$\{T \leq t\} = \bigcap_{n \geq 1} \left\{T < t + \tfrac{1}{n}\right\}.$$

Since $\mathcal{F}^+_{t+\frac{1}{n}} \subseteq \mathcal{F}_{t+\varepsilon}$ for all $\varepsilon > 0$, we get $\{T \leq t\} \in \bigcap_{\varepsilon > 0} \mathcal{F}^+_{t+\varepsilon} = \mathcal{F}_t^+$ (using right continuity). □

10.4 Blumenthal's 0/1 Law

Q: For each $t \geq 0$, $\mathcal{F}_t^0 \subseteq \mathcal{F}_t^+$. Does $\mathcal{F}_t^+$ really contain more "information"?
A: Not really!

Theorem 10.6. Blumenthal's 0/1 Law

Let $\{B_t : t \geq 0\} \sim$ sBM and let $A \in \mathcal{F}_0^+$. Then $\mathbb{P}(A) \in \{0, 1\}$.

Proof. Since $\{B_t : t \geq 0\} \perp\!\!\!\perp \mathcal{F}_0^+$ (by the Markov property with $s = 0$), we have: $A \in \mathcal{F}_0^+$ implies $A = \varphi(\{B_t : t \geq 0\})$ for some measurable φ. Therefore $A \perp\!\!\!\perp A$, which

gives

$$\mathbb{P}(A) = (\mathbb{P}(A))^2 \quad \Longrightarrow \quad \mathbb{P}(A) \in \{0, 1\}. \qquad \square$$

Q: What does Blumenthal's 0/1 law mean for sBM?

Lemma 10.7. Instantaneous Behavior of BM

Let $\{B_t : t \geq 0\} \sim$ sBM. Define

$$T = \inf\{t > 0 : B_t > 0\}, \qquad \sigma = \inf\{t > 0 : B_t = 0\}.$$

Then $\mathbb{P}(T = 0) = \mathbb{P}(\sigma = 0) = 1$.

Proof. $\mathbb{P}(T = 0) = 1$**:** We have

$$\{T = 0\} = \bigcap_{n \geq 1} \left\{\exists\, \varepsilon \in \left(0, \tfrac{1}{n}\right) \text{ s.t. } B_\varepsilon > 0\right\} \in \mathcal{F}_0^+.$$

By Blumenthal's 0/1 law, $\mathbb{P}(T = 0) \in \{0, 1\}$.
For any $t > 0$, $\mathbb{P}(T \leq t) \geq \mathbb{P}(B_t > 0) = \frac{1}{2}$. Therefore $\mathbb{P}(T = 0) = 1$.

$\mathbb{P}(T' = 0) = 1$**:** By the same argument applied to $T' = \inf\{t > 0 : B_t < 0\}$, we get $\mathbb{P}(T' = 0) = 1$.

$\mathbb{P}(\sigma = 0) = 1$**:** Since $\mathbb{P}(T = 0) = \mathbb{P}(T' = 0) = 1$, BM is immediately positive and immediately negative after time 0. By the intermediate value theorem (and continuity of paths), BM must hit 0 in every interval $(0, \varepsilon)$. Hence $\mathbb{P}(\sigma = 0) = 1$. $\square$

10.5 Strong Markov Property

So far: BM refreshes after a fixed time. What about random times?
A: Not true in general. True for stopping times.

Definition 10.8. Stopped σ-Algebra

Let T be a stopping time with respect to $\{\mathcal{F}_t^+ : t \geq 0\}$. The *stopped σ-algebra* is

$$\mathcal{F}_T^+ = \{A \in \sigma(\{B_t : t \geq 0\}) : A \cap \{T < t\} \in \mathcal{F}_t^+ \ \forall\, t \geq 0\}.$$

Theorem 10.9. Strong Markov Property

Let $\{B_t : t \geq 0\} \sim$ sBM and T be a stopping time with respect to $\{\mathcal{F}_t^+ : t \geq 0\}$ with $\mathbb{P}(T < \infty) = 1$. Set $W_t = B_{T+t} - B_T$. Then $\{W_t : t \geq 0\}$ is a standard Brownian motion independent of $\mathcal{F}_T^+$.

Remark 10.10. Monotonicity of Stopped σ-Algebras

If T, S are stopping times with $T \leq S$, then $\mathcal{F}_T^+ \subseteq \mathcal{F}_S^+$.

Proof. **Step 1: Countable stopping times.** Assume T takes countably many values $0 \leq t_1 \leq t_2 \leq \cdots$. Define $W_t^i = B_{t+t_i} - B_{t_i}$.

Fix $A \in \mathcal{F}_T^+$ and $E \in \mathcal{B}_C([0,\infty))$. Then

$$\begin{aligned}\mathbb{P}[\{W_t : t \geq 0\} \in E,\ A] &= \sum_{i=1}^{\infty} \mathbb{P}[\{W_t^i : t \geq 0\} \in E,\ A \cap \{T = t_i\}] \\ &= \sum_{i=1}^{\infty} \mathbb{P}[\{W_t^i : t \geq 0\} \in E] \cdot \mathbb{P}[A \cap \{T = t_i\}]\end{aligned}$$

where the second equality uses $\{W_t^i : t \geq 0\} \perp\!\!\!\perp \mathcal{F}_{t_i}^+$ and $A \cap \{T = t_i\} \in \mathcal{F}_{t_i}^+$.
Since $\{W_t^i : t \geq 0\}$ is sBM for each i, this equals

$$= \mathbb{P}(A) \cdot \mathbb{P}(\{B_t : t \geq 0\} \in E).$$

Step 2: General stopping times. For a general stopping time T, define the dyadic approximation

$$T_n = \frac{m+1}{2^n} \quad \text{if } \frac{m}{2^n} \leq T < \frac{m+1}{2^n}.$$

Check: $\{T_n : n \geq 1\}$ is a stopping time with respect to $\mathcal{F}_t^+$. Indeed, $\{T_n < t\} = \bigcup_{m=0}^{\lfloor 2^n t \rfloor} \{T_n = \frac{m}{2^n}\} \in \mathcal{F}_t^+$.
Define $W_t^{(n)} = B_{T_n+t} - B_{T_n}$. By Step 1, $\{W_t^{(n)} : t \geq 0\} \perp\!\!\!\perp \mathcal{F}_{T_n}^+$ for all $n \geq 1$.
Since $T_n \geq T$, we have $\mathcal{F}_T^+ \subseteq \mathcal{F}_{T_n}^+$, so $\{W_t^{(n)} : t \geq 0\} \perp\!\!\!\perp \mathcal{F}_T^+$ for all $n \geq 1$.
As $n \to \infty$, $T_n \downarrow T$, so $W_t^{(n)} \to W_t$ almost surely (by continuity of paths). Therefore $\{W_t : t \geq 0\} \perp\!\!\!\perp \mathcal{F}_T^+$. □

10.6 Application: Running Maximum is Not a Stopping Time

Lemma 10.11. Running Maximum Time is Not a Stopping Time

Let $T = \inf\{t : B_t = \max_{s\in[0,1]} B_s\}$. Then T is *not* a stopping time.

Proof. First, observe $\mathbb{P}(T < 1) = 1$.
Define $W_t = B_{1-t} - B_1$ for $0 \leq t \leq 1$. Then $\{W_t : t \in [0,1]\} \sim$ sBM on $[0,1]$.
If $\{T = 1\}$, then $W_t < 0$ in a neighborhood of zero (since B_t achieves its maximum at time 1). Formally, $\rightarrow\leftarrow$.
Now consider $\{B_{T+t} - B_T : t \geq 0\}$. By the strong Markov property (if T were a stopping time), this would be a sBM. But $B_{T+t} - B_T \leq 0$ for small $t > 0$ (since B_T is the maximum of B on $[0,1]$), which contradicts the fact that a sBM is immediately positive (by Definition 10.7). This is a contradiction! □

Chapter 11

Reflection Principle, Continuous-Time Martingales

11.1 Reflection Principle for Brownian Motion

Let $a > 0$ and define $T = \inf\{t \geq 0 : B_t > a\}$.

Remark 11.1. T is a Stopping Time

Observation 1: T is a stopping time with respect to $\mathcal{F}_t^+$.
Indeed,

$$\{T \leq t\} = \bigcap_{s>t} \{T < s\} = \bigcap_{s>t} \bigcup_{n \in (0,s) \cap \mathbb{Q}} \{B_n > a\} \in \mathcal{F}_t^+.$$

Remark 11.2. T is Finite Almost Surely

Observation 2: $\mathbb{P}(T < \infty) = 1$.
This follows from $\limsup_{n\to\infty} \frac{B_n}{\sqrt{n}} = +\infty$ with probability 1.

Definition 11.3. Reflected Brownian Motion

Define the *reflected* process

$$\tilde{B}_t = \begin{cases} B_t & \text{if } t \leq T, \\ 2B_T - B_t & \text{if } t > T. \end{cases}$$

Lemma 11.4. Reflected Process is sBM

$\{\tilde{B}_t : t \geq 0\}$ is a standard Brownian motion.

Proof. Define $W_t = B_{T+t} - B_T$ and $U_t = B_{t \wedge T}$.
Consider the map $\varphi \colon C([0,\infty)) \times [0,\infty) \times C([0,\infty)) \to \mathbb{R}$ defined by

$$\varphi(U, T, W) = \begin{cases} U_t & \text{if } t \leq T, \\ a + W_{t-T} & \text{if } t > T. \end{cases}$$

Then $B_t = \varphi(U, T, W)$ and $\tilde{B}_t = \varphi(U, T, -W)$.

Claim 11.5. (U, T) is $\mathcal{F}_T^+$-Measurable

The pair (U, T) is $\mathcal{F}_T^+$-measurable. (Check: $U_t = B_{t\wedge T}$ is $\mathcal{F}_T^+$-measurable.)

By the strong Markov property, $\{W_t : t \geq 0\} \perp\!\!\!\perp \mathcal{F}_T^+$. Since $-W$ has the same distribution as W (by symmetry of sBM), we get

$$\varphi(U, T, W) \overset{d}{=} \varphi(U, T, -W).$$

Therefore $\{\tilde{B}_t : t \geq 0\} \sim$ sBM. □

11.2 Distribution of the Running Maximum

Theorem 11.6. Distribution of the Running Maximum

Let $M_t = \max_{0\leq s\leq t} B_s$. Fix $a > 0$ and let $T = \inf\{t \geq 0 : B_t > a\}$. Then

$$\mathbb{P}[M_t \geq a] = 2\,\mathbb{P}[B_t \geq a].$$

Consequently, $M_t \overset{d}{=} |Z|$ where $Z \sim N(0, t)$.

Proof. Decompose the event:

$$\begin{aligned}\mathbb{P}[M_t \geq a] &= \mathbb{P}[M_t \geq a,\ B_t \geq a] + \mathbb{P}[M_t \geq a,\ B_t < a] \\ &= \mathbb{P}[B_t \geq a] + \mathbb{P}[M_t \geq a,\ \tilde{B}_t > a],\end{aligned}$$

where the second term uses the reflected Brownian motion $\tilde{B}$: on the event $\{M_t \geq a,\ B_t < a\}$, we have $T \leq t$ and $\tilde{B}_t = 2B_T - B_t = 2a - B_t > a$.
Since $\{\tilde{B}_t : t \geq 0\}$ is also a sBM,

$$\mathbb{P}[M_t \geq a,\ \tilde{B}_t > a] = \mathbb{P}[\tilde{B}_t > a] = \mathbb{P}[B_t > a] = \mathbb{P}[B_t \geq a].$$

Therefore

$$\mathbb{P}[M_t \geq a] = \mathbb{P}[B_t \geq a] + \mathbb{P}[B_t \geq a] = 2\,\mathbb{P}[B_t \geq a].$$

Since $B_t \sim N(0, t)$, we have $\mathbb{P}[M_t \geq a] = 2\,\mathbb{P}[B_t \geq a] = \mathbb{P}[|B_t| \geq a]$, so $M_t \overset{d}{=} |Z|$ where $Z \sim N(0, t)$. □

11.3 Continuous-Time Martingales

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with filtration $\{\mathcal{F}_t : t \geq 0\}$, and let $\{X_t : t \geq 0\}$ be a stochastic process adapted to $\{\mathcal{F}_t : t \geq 0\}$.

Definition 11.7. Continuous-Time Martingale

$\{(X_t, \mathcal{F}_t) : t \geq 0\}$ is a *martingale* if:

(i) $\mathbb{E}|X_t| < \infty$ for all $t \geq 0$.

(ii) $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ for all $s \leq t$.

Example 11.8. Brownian Motion is a Martingale

Let $\{B_t : t \geq 0\} \sim$ sBM with the canonical right-continuous filtration $\{\mathcal{F}_t^+ : t \geq 0\}$. Then $\{(B_t, \mathcal{F}_t^+) : t \geq 0\}$ is a martingale.

Proof. By the Markov property, $B_t - B_s \perp\!\!\!\perp \mathcal{F}_s^+$. Therefore

$$\mathbb{E}[B_t \mid \mathcal{F}_s^+] = \mathbb{E}[B_t - B_s \mid \mathcal{F}_s^+] + B_s = \mathbb{E}[B_t - B_s] + B_s = 0 + B_s = B_s. \qquad \square$$